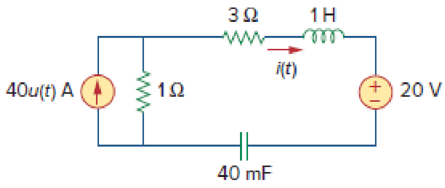
## Chapter 16, Problem 39P

### Problem

For the network in Fig. 16.62, find $i(t)$ for $t > 0$.

**Figure 16.62:**




### Step-by-step solution

**Step 1** of 5

Refer to circuit diagram in Figure 16.62 in the textbook.

It is known that,

$$u(t)=\begin{cases}0 & t<0\\1 & t\geq 0\end{cases}$$

For $t<0$, the value of current source magnitude is zero. Capacitor acts as open circuit and inductor acts as short circuit.

The modified circuit is shown in Figure 1.

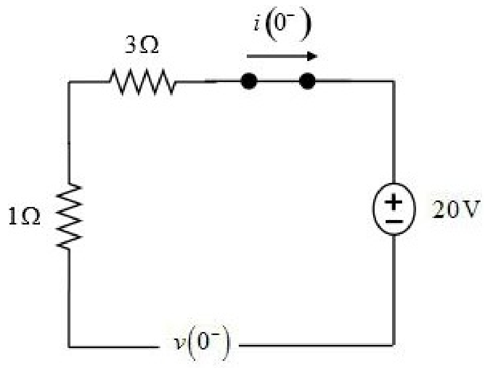



Figure 1

**Step 2** of 5

The circuit is not a closed circuit, the current through the inductor is 0 A.

$$i(0^-)=0\,\text{A}$$

The voltage across the capacitor for $t<0$ is,

$$v(0^-)=20\,\text{V}$$

**Step 3** of 5

For $t > 0$, the s-domain circuit is shown in Figure 2.

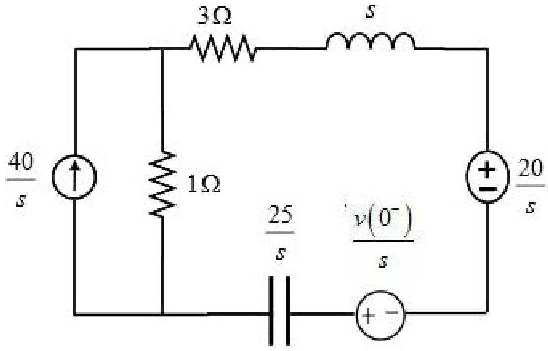



Figure 2

**Step 4** of 5

Replace current source in parallel with resistance by an equivalent voltage source in series with the same resistance. The modified circuit is shown in Figure 3.

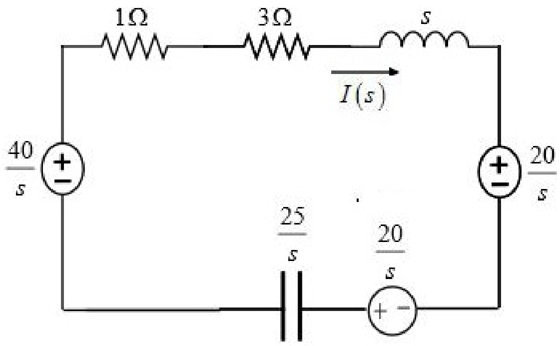



Figure 3

**Step 5** of 5

Apply Kirchhoff voltage law.

$$-\frac{40}{s}+(1+3)I(s)+sI(s)+\frac{20}{s}-\frac{20}{s}+\frac{25}{s}I(s)=0$$

$$I(s)\left[s+4+\frac{25}{s}\right]=\frac{40}{s}$$

$$I(s)\left[s^2+4s+25\right]=40$$

$$\begin{aligned} I(s) &= \frac{40}{s^2+4s+25} \\ &= \frac{40}{(s+2)^2+21} \\ &= \frac{(8.733)(4.58)}{(s+2)^2+4.58^2} \end{aligned}$$

It is known that,

$$L^{-1}\left(\frac{b}{(s+a)^2+b^2}\right)=e^{-at}\sin(bt)u(t)$$

Apply Inverse Laplace transform.

$$i(t)=8.733e^{-2t}\sin(4.58t)\text{ A}$$

Therefore, the current, $i(t)$ for $t > 0$ is $\boxed{8.733e^{-2t}\sin(4.58t)\text{ A}}$.